Inference Rules for Functional Dependency

Suppose $F = \{ Emp \rightarrow Dept, Dept \rightarrow Dmanager \}$, then the closure of the functional dependency F is denoted by F^+ i.e., it is the set of all dependencies that include F as well as all dependencies that can be inferred / derived from the functional dependency F.

Armstrong's Axioms

- IR_1 (Reflexive rule): If $X \supseteq Y$, then $X \rightarrow Y$
- IR_2 (Augmentation Rule): If $\{X \rightarrow Y\}$ then $XZ \rightarrow YZ$
- IR_3 (Transitive Rule): If $\{X \rightarrow Y, Y \rightarrow Z\}$ then $X \rightarrow Z$

IR_4 (Decomposition or Projective Rule): If $\{X \rightarrow YZ\}$ then $X \rightarrow Y$

IR_5 (Union or Additive Rule): If $\{X \rightarrow Y, X \rightarrow Z\}$ then $X \rightarrow YZ$

IR_6 (Pseudo transitive Rule): If $\{X \rightarrow Y, WY \rightarrow Z\}$, then $WX \rightarrow Z$

Reflexive Rule states that a set of attributes always determines itself or any of its subset.

Given $X \supseteq Y$. If t_1 and t_2 are two tuples present in R then $t_1[X] = t_2[X]$ if
$t_1[Y] = t_2[Y]$
as $X \supseteq Y$
So $X \rightarrow Y$

NOTE

A functional dependency X, Y is said to be trivial, if $X \supseteq Y$, otherwise it is not trivial.

IR_2 (Augmentating rule): If $\{X \to Y\}$ then $XZ \to YZ$

Augmentation Rules state that adding same set of attributes to both the left and Right hand side of a dependency results in another valid dependency

Given $X \to Y$

Assume that $XZ \to YZ$ does not hold.

$t_1[X] = t_2[X]$ — ①

$t_1[Y] = t_2[Y]$ — ②

$t_1[XZ] = t_2[XZ]$ — ③

$t_1[YZ] \neq t_2[YZ]$ — ④

From equ^ ① and ③

$t_1[Z] = t_2[Z]$ — ⑤

From equ^ ② & ⑤

$t_1[YZ] = t_2[YZ]$ — ⑥ contradicts equ^ ④

Thus, there exist $XZ \to YZ$ proved.

IR_3 (Transitive Rule): If $X \to Y$, $Y \to Z$ then $X \to Z$

Given $X \to Y$ and $Y \to Z$

$t_1[X] = t_2[X]$ and $t_1[Y] = t_2[Y]$ — ①

$t_1[Y] = t_2[Y]$ and $t_1[Z] = t_2[Z]$ — ②

From equ^ ① & ②

$t_1[X] = t_2[X]$

$t_1[Z] = t_2[Z]$

So, $X \to Z$ proved.

IR_4 (Decomposition or projection Rule): If $X \to YZ$, then $X \to Y$

We can remove attributes from the right hand side of a dependency

Given: $X \to YZ$ — ①

$YZ \to Y$ — ② (Using IR_1 $YZ \supseteq Y$)

$X \to Y$ (Using IR_3) proved.

IR_5 (Using Arbitrative Rule) : If $x \to y$, $x \to z$, then $x \to yz$

Given : $x \to y$ — ①

$x \to z$ — ②

$xx \to xy$ (Using IR_2, augmenting x we will get $x \to xy$) — ③

from ③ & ① $xy \to yz$ (using IR_2) — ④

$x \to yz$ proved. (using IR_3)

IR_6 (Pseudo transitive Rule) : If $x \to y$, $wy \to z$, then $wx \to z$

Given $x \to y$ — ①

$wy \to z$ — ②

from eqn ①

$wx \to wy$ (using IR_2) — ③

from eqn ③ & ②

$wx \to z$ (using IR_3) proved.